NEXUS TECHNOLOGY CUP 2026 · COLLEGE TRACK · UNIVERSITY OF LOUISIANA MONROE

Plumbline

An assistant for blind hands that cannot tell you something it has not measured.

When a blind person tests an electronic circuit, Plumbline works out what is broken *and* where their hand really touched, from the reading alone. So it catches the slip instead of confidently blaming the wrong part.

21×
fewer confidently wrong diagnoses than the published method (team simulation)

99%
of "your probe isn't where you think" warnings were correct

35 mi
to the Louisiana Center for the Blind in Ruston, about 30 blind adults in daily shop training

Jeevan Parajuli and Pankaj Bhatta · Results from the team's simulation (1,500 faults × 5 seeds); real-user validation is the next step.

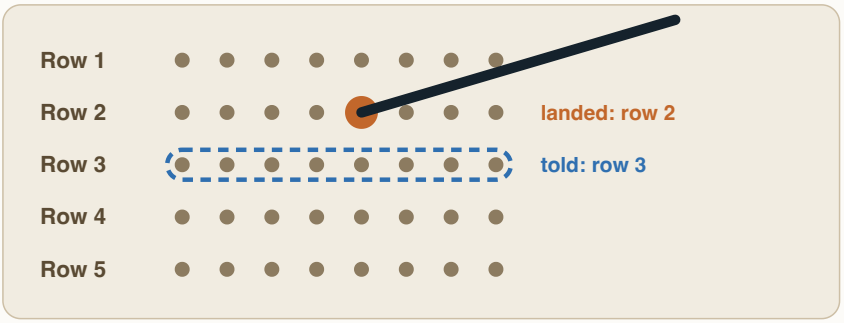
What's inside

1. **The idea**: the problem, in plain words and in blind users' own words
2. **The solution**: how Plumbline works, step by step
3. **The proof so far**: results and the 20-second worked example
4. **Why it's new**: what exists, and the exact gap
5. **Why this can win**: mapped to the four Nexus judging criteria
6. **Risks and how to fix them**: before a judge finds them
7. **The plan**: what to do before the application, and before the December finals
8. **Hard questions**: and strong answers

1 · THE IDEA

A blind person can hear the meter, but can’t see the probe

To find a broken part in a circuit, you touch a meter probe to points on a breadboard (a board of numbered rows of holes) and read the voltage. Talking multimeters read the number aloud, so the *display* problem is solved. The *probe* problem never was.



Talking meter says

“3.671 volts”

The number is right.
The location is wrong.

Every existing tool assumes row 3.

The core failure: the measurement is correct, but it came from a different spot than the one the system asked for. A sighted person would notice instantly. A blind person counting holes by touch cannot.

WHY THIS BREAKS TODAY’S TOOLS

Every troubleshooting method, from textbook checklists to published research algorithms, assumes that the point you were asked to measure is the point you measured. When the hand slips one row, the reading gets interpreted as if it came from the right row. The tool then names the wrong broken part with high confidence. A blind user has no way to tell that it’s wrong.

BLIND USERS DESCRIBE EXACTLY THIS PROBLEM

> “sometimes I wonder whether or not the inaccuracy is really more my picture taking skills”
>
> Participant 23, “Misfitting With AI,” University of Michigan, ASSETS 2024 (26 blind adults via the National Federation of the Blind)

She can’t tell whether an error is hers or the machine’s. The same study calls the extra checking blind users do “epistemic labor.”

> “the only way to make sure that it was 100% correct.”
>
> Blind participant breaking a task into many small questions, “Say It My Way,” University of Maryland, CHI 2026 (11 participants, 418 interactions with Be My AI)

She invented step-by-step testing by hand because nothing does it for her. Plumbline automates that, and picks the questions optimally.

> “He can’t see the probes, so… I can’t imagine this going well.”
>
> EEVblog forum, thread about a blind engineer needing a talking multimeter. The blind engineer counted chip pins and connectors by feel.

AND GENERAL AI CAN’T SIMPLY SOLVE IT

The CircuitSense benchmark (arXiv 2509.22339, 8,006 circuit problems) found frontier models recognize circuit parts almost perfectly but fail to reason about them. Gemini-2.5-Pro reached 19% on derivations, Claude-Sonnet-4 about 6%, GPT-4o about 5%. Seeing isn’t understanding, and a chatbot also can’t know where a blind user’s hand is.

Sources: arXiv 2408.06546 (ASSETS 2024) · DOI 10.1145/3772318.3791834 (CHI 2026) · arXiv 2602.18623 (Stony Brook 2026) · TangibleCircuits, CHI 2020 · arXiv 2509.22339 (CircuitSense). Open each before quoting.

2 · THE SOLUTION

Plumbline solves for the fault and the hand at the same time

Every point on a circuit has a different expected voltage, and each possible broken part changes those voltages in its own way. So a reading carries a fingerprint of *which point was actually touched*, as well as *what is broken*. Plumbline uses both.



The loop repeats until one fault is clearly most likely. If two faults truly can't be told apart by voltages, Plumbline says so instead of guessing.

HOW IT WORKS, IN PLAIN WORDS

1. **Know the circuit.** A circuit solver (modified nodal analysis) computes the expected voltage at every row for every possible fault. The current circuit has 33 possible faults.
2. **Expect the hand to slip.** A small model of touch accuracy: right row about 78% of the time, one row off about 8% each way, two rows off about 3% each way.
3. **Read the evidence both ways.** For each reading, ask: which fault, and which actual row, best explain this number?
4. **Choose the next row wisely.** Pick the row whose reading will shrink uncertainty the most (information gain).
5. **Speak up about the hand.** If the reading most likely came from a different row, say so out loud.
6. **Refuse to guess.** Only announce a fault when confident; report genuinely indistinguishable faults as unresolved.

THE KEY INSIGHT

> Probe position is recoverable exactly as far as it matters, and no further. Where the system can't tell two rows apart, their voltages match, so it doesn't need to.

WHAT THE USER EXPERIENCES

- A voice says which row to touch.
- They touch it and say the number the talking meter reads.
- The voice either asks for the next row, warns "I think your hand is on row 2," or names the broken part and how sure it is.
- Hardware needed: a breadboard, a battery, a cheap talking meter and a phone or laptop.

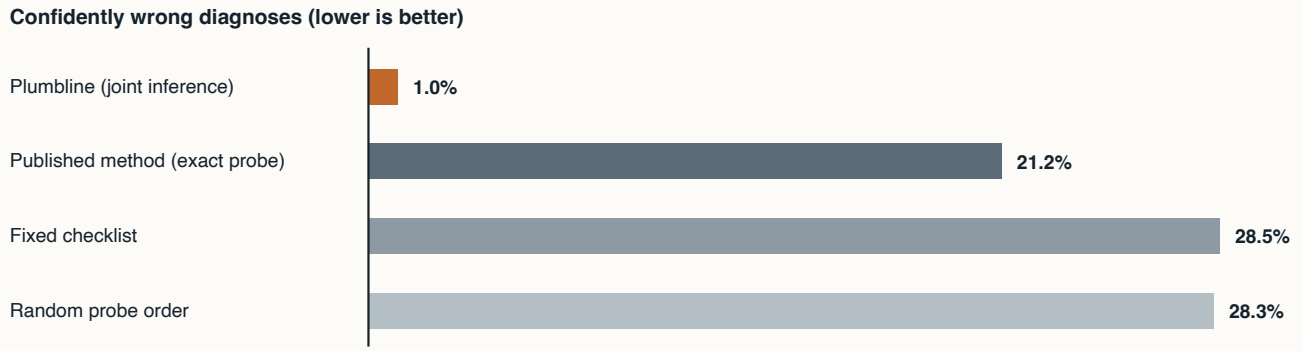
WHAT IS BUILT TODAY

The engine, the circuit solver, the 33-fault catalog, the comparison experiments and figures (files: circuit.py, engine.py, experiment.py, stability.py, mkfigs.py).

3 · THE PROOF SO FAR

21× fewer confidently wrong answers

Same circuit, same meter, same simulated hands; 1,500 simulated faults per method across five random seeds. “Confidently wrong” means the method named the wrong broken part while claiming over 90% confidence.



Team simulation. Correct diagnosis rates: Plumbline 99.0% ± 0.4, published method 78.7% ± 1.8, checklist 69.8% ± 2.9, random 70.2% ± 1.1. Plumbline also issued 178 “your probe isn’t where you think” warnings, and 176 were correct (99%).

THE 20-SECOND WORKED EXAMPLE (ALSO THE DEMO)

STEP	PUBLISHED METHOD	PLUMBLINE
True fault	Resistor R3 is ten times too small. Both are told to probe row 3. Both hands land on row 2.	
First reading: 3.671 V	Immediately **100% sure R2 is shorted**. Wrong, and certain.	Leans toward a wrong candidate (63.5%) but **not confident**, so it asks for another reading.
Second reading: 0.338 V	Already finished, with the wrong answer.	**R3 is ten times too small, 97.6% sure.** Correct.

Honest status. These numbers come from the team’s simulation, which uses an assumed model of how often hands slip. The team implemented both Plumbline and the baseline. No blind person has used Plumbline yet. Section 6 shows how to make the result bulletproof, and section 7 how to replace the assumption with real measurements.

ROBUST TO THE ASSUMPTION

The team swept the hand accuracy from 60% to 100%. Plumbline stayed under 2% confidently wrong across the whole range, while the other methods climbed past 35%. Whatever the real accuracy turns out to be, the conclusion holds.

4 · WHY IT'S NEW

Everyone models noisy tests. Nobody models a noisy hand.

Say the prior work first. That is what makes the novelty claim believable to a technical judge.

WHAT EXISTS	WHAT IT DOES	WHAT IT CAN'T DO
Entropy-based test-point selection (Gao et al., 2014)	Picks the most informative point to measure in analog circuits	Assumes the measured point is the requested point
Sequential diagnosis with unreliable tests (incl. a POMDP treatment, IEEE 2019)	Handles tests that sometimes give wrong results	Models a noisy *result*, not a test run on the wrong point
Model-based diagnosis (de Kleer & Williams, 1980s)	Reasons from a circuit model to the fault	Same assumption: the probe is where you think
Talking multimeters	Read the number aloud	No reasoning; can't know where the probe is
TangibleCircuits (CHI 2020), AccessibleCircuits (CHI 2021)	Tactile models and parts to understand circuits	Don't touch measurement or diagnosis
Be My AI, ChatGPT and similar	Describe and answer questions	Can't know the hand's position; weak at circuit reasoning (CircuitSense)
Plumbline	**Infers fault and true probe position jointly**	**Catches the slip, recovers, and refuses to guess**

THE PRECISE TECHNICAL DELTA

In control-engineering terms, the field models **sensor noise**. Plumbline models **actuator noise with a recoverable hidden state**: the person is the actuator, the probe's true row is the hidden state, and it is recoverable because different rows carry different voltages.

The accessibility constraint is what creates the research problem. For a sighted user the hand isn't noisy, so the problem never appears, which is why it is still open in 2026.

THE MATH, IN FOUR LINES

```
Likelihood: P(r | f, n) = Σ_l P(l | n) · Normal(r ; V_f(l), σ)
Update: b′(f) ∝ b(f) · P(r | f, n)
Next row: n* = argmax_n [ H(b) − E_r H(b′ | r) ]
Warn if: P(l ≠ n | r) > 0.5
```

f = fault, n = instructed row, l = actual row, r = reading, V_f(l) = voltage on row l under fault f (from the circuit solver), σ = meter noise (0.02 V), H = entropy.

5 · WHY THIS CAN WIN

It is strong on all four official Nexus criteria

Nexus judges on Impact, Degree of difficulty, Novelty and Feasibility, and insists on a working demonstration. Here is how Plumbline answers each.

Impact · a real, documented gap

Three studies with 51 blind participants show blind users cannot verify what AI tells them. Plumbline is a concrete fix: an assistant that only states what it measured and accounts for human error. The same engine extends to any measuring tool (thermometer, scale, caliper, pH meter).

Difficulty · real math, not a wrapper

Joint inference over fault and hidden probe position, with a circuit solver inside an information-gain loop. Not an API call or a prompt. The result appeared only once the math was right.

Novelty · a precise, defensible gap

The closest prior work is named and credited; the remaining gap (uncertainty in *which* test ran) is specific and checkable. A measured 21× improvement over the published baseline is the strongest form of novelty claim.

Feasibility · it already runs

The engine works and reproduces. The demo needs a breadboard, a 9 V battery and a phone. No regulatory path, no IRB needed for the October demo, nothing to manufacture.

WHY IT BEATS THE TYPICAL COLLEGE ENTRY

WHAT JUDGES USUALLY SEE	WHAT PLUMBLINE SHOWS INSTEAD
A slide deck about what could exist	A working engine and a reproducible experiment
An LLM wrapper or chatbot	Probabilistic inference with a physics model inside
“Nobody does this” without proof	Three named prior methods and the exact gap they leave
Accuracy claims with no baseline	Four methods compared, 1,500 trials, five seeds
A demo on a screen	A live, blindfolded run where the system catches a deliberate slip

COMPARED WITH 2025’S WINNERS

- **FarmMind (College 1st):** a specialized workflow with credible grounding. Plumbline has the same pattern, plus a harder technical core and a measured result.
- **Dristi (College 3rd, also ULM):** AI scene description for blind users. Plumbline is clearly different: it doesn’t describe, it reasons about measurements and catches errors.
- **Encore CO_2 (Open 1st + Most Innovative):** hard science made tangible. Plumbline’s blindfolded live demo is that kind of moment.

OUR INTERNAL SCORE (NOT THE NEXUS RUBRIC)

STAGE	IMPACT	DIFFICULTY	NOVELTY	FEASIBILITY	DEMO	OVERALL
Today	3.5	4	4.5	5	5	**~87**
After the fixes in section 6 and real data from Ruston	4	4.5	4.5	5	5	**~91**

0–5 per criterion; College = 5I+5D+4N+4F+2Demo; Most Innovative = 9N+7D+4Demo; Overall = 60/40 blend. This ties our best research idea's ceiling but is far more likely to reach it, because the core already works and real users are 35 miles away.

Say these before a judge does

WEAKNESS	WHY A JUDGE MIGHT PUSH	FIX
The 21× could look circular	The simulation injects hand slips using the same model Plumbline assumes, against a baseline that assumes no slips.	Test mismatch: keep Plumbline's model fixed and simulate different true behavior (lopsided slips, 3+ row jumps, poor contact reading nothing, touching two rows). Report all results.
Baselines too easy	"Why not just measure twice?"	Add "re-measure until two readings agree," a standard outlier-tolerant method, and a real ChatGPT/Claude run on the same netlist. Don't claim what ChatGPT says until it's measured.
No blind person has used it	All human evidence is from other people's studies.	The Louisiana Center for the Blind visit (section 7). Even one real session changes the story.
Small market	Blind people debugging breadboards is a small group.	Show the same engine on a second instrument (e.g., a kitchen or lab measurement) to prove it generalizes. Pitch as "trustworthy measurement assistance," with electronics as the wedge.
Toy-sized circuit	6 rows and 33 faults may look small.	Lead with the formulation; show it scaling to a bigger circuit; name multiple simultaneous faults as the next frontier.
Single faults, DC only	Real circuits have more going on.	Say "deliberately narrow wedge." Single-fault DC is where the method is proven first.
The kill test	If blind users place probes over 95% accurately, the advantage shrinks.	Run this test first at the Center. Either answer is publishable: nobody has measured blind probe placement before.

Language to use: "21× fewer confidently wrong diagnoses," not "fewer lies." Measured, specific claims build trust with technical judges.

Before the application, and before December

BEFORE THE APPLICATION (TUE OCT 27)

1. **Stress tests:** mismatched hand models, poor contact and two-row bridging, plus the stronger baselines.
2. **Voice loop:** wrap the engine in speech (browser Web Speech API). The user speaks the reading.
3. **Real breadboard:** build the exact circuit from circuit.py with a 9 V battery and an LED.
4. **Record on camera:** blindfolded run; then a deliberate off-by-one slip so the warning fires. This is the key video.
5. **Screen-record experiment.py** producing the numbers, and take high-resolution prototype photos.
6. **Email Pam Allen**, Executive Director of the Louisiana Center for the Blind, asking for help, not "a study."
7. **Confirm a ULM faculty advisor** (required for the College Track).
8. **Open every cited paper** and confirm each quote and number.

Don't build: accounts, databases, dashboards, mobile apps or cloud infrastructure. Every hour there is an hour not spent on what is actually novel.

BEFORE THE FINALS (DEC 14–16)

1. **Measure real hand accuracy** at the Louisiana Center for the Blind, 101 South Trenton Street, Ruston: 8 participants × 20 placements = 160 data points. This is the first published tactile probe-placement data for blind users.
2. **Re-run the experiments** with the measured numbers.
3. **Sessions with blind users** finding real faults; record success, time and quotes.
4. **One letter of support** from an instructor or rehabilitation professional.
5. **Extend** to a larger circuit and one second instrument.
6. **Rehearse** the 10-minute live demo with the Nexus coach.

THE 10-MINUTE PITCH

1. Play an AI confidently describing something wrong. Ask: how would you know?
2. Participant 23: "is the mistake mine or the machine's?"
3. Show the circuit: one LED, six rows.
4. Blindfolded, live. Asked for row 3, land on row 2 on purpose.
5. Show what the standard approach does with that reading.
6. Plumbline: "that reading did not come from row 3," then the right answer.
7. The robustness figure: the published method is fine for people who can see.
8. 21×, 1,500 trials, five seeds, reproducible.
9. Ruston: real blind users, real measured data.
10. "It cannot tell you something it has not measured."

Five questions judges will ask

QUESTION	STRONG ANSWER
“Isn’t the 21× just because you simulated the noise your method assumes?”	“We tested that. When the true hand behavior differs from our model, including lopsided slips, big jumps and bad contact, Plumbline still [result]. And across hand accuracy from 60% to 100%, it stays under 2% confidently wrong.” *Run the mismatch tests first so this answer is real.*
“Why not just ask the user to measure twice?”	“We compared that baseline: [result]. Re-measuring catches random slips but not systematic ones, like always counting from the wrong end. And it doubles the work without telling the user what went wrong.”
“Hasn’t sequential fault diagnosis been done for decades?”	“Yes: Gao 2014, the IEEE 2019 POMDP work, de Kleer and Williams. All of them model a noisy test result. None model a test run on the wrong point and recover from it. That only becomes a problem when the user can’t see their hand.”
“How many blind people actually need this?”	“Electronics is the wedge: blind STEM students, makers and technicians. The principle is trustworthy measurement for anyone who can’t visually check their tool, and the same engine works with any instrument. We’re validating with students at the Louisiana Center for the Blind.”
“Has a blind person used it?”	Today: “Not yet. That’s our next step, and we’ve contacted the Louisiana Center for the Blind in Ruston.” By December: “Yes, [N] participants, measured hand accuracy [X]%, and here’s what they said.”

The display was solved years ago. The probe never was. Plumbline is the first assistant that knows your hand might be wrong, and can tell you when it is.

Prepared Oct 25, 2026 from the team’s Plumbline build brief. All performance numbers come from the team’s simulation code (circuit.py, engine.py, experiment.py, stability.py) and have not yet been independently reproduced. Bracketed [result] values must be filled with measured results only.